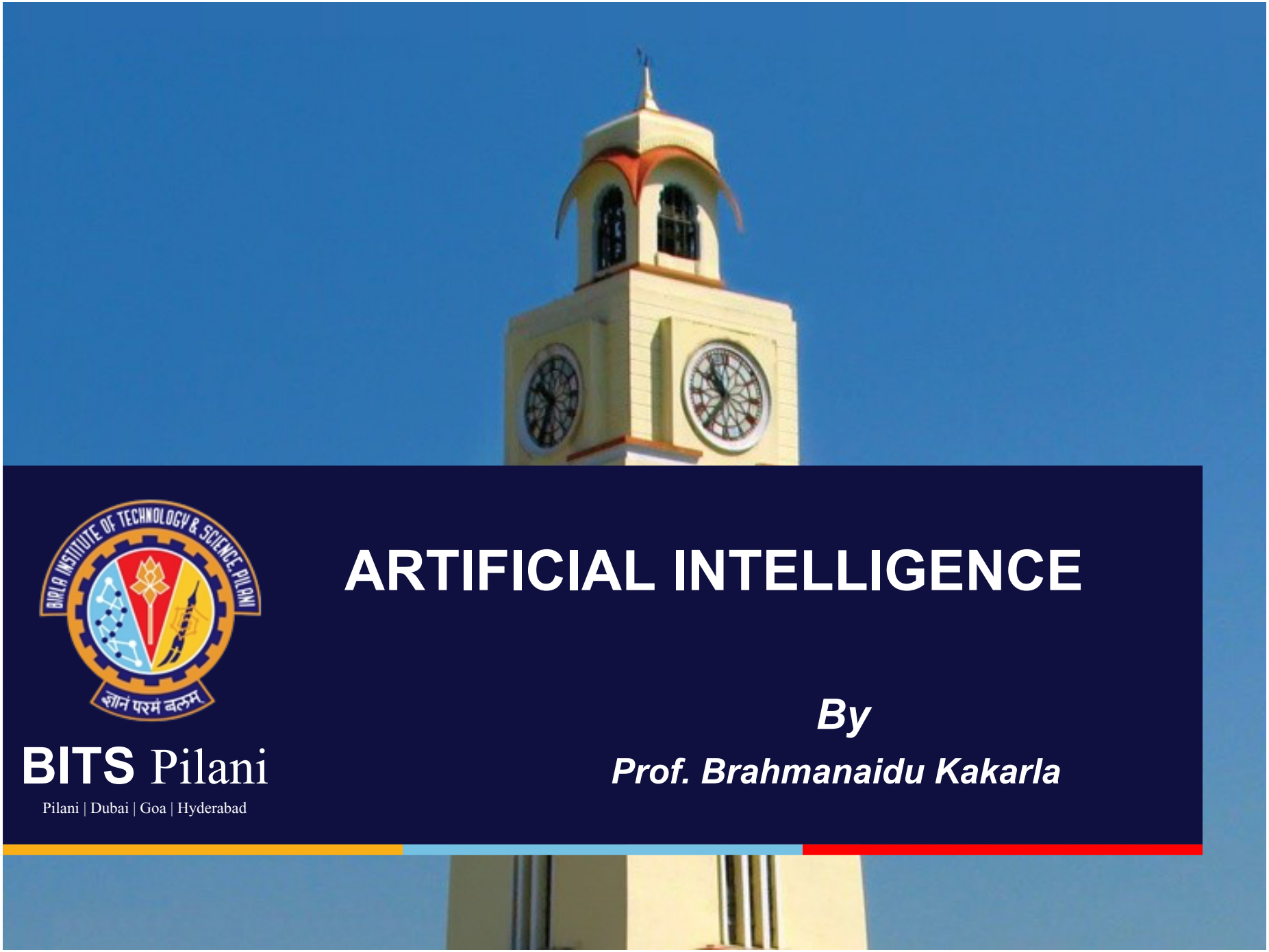


ARTIFICIAL INTELLIGENCE

By

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Contact Session-6

Probability, Bayesian Networks and Hidden Markov Models

Uncertainty

- You can reach Bangalore Airport from MG Road within 90 m ins if you go by route A.
  - There is uncertainty in this information due to partial observability and non determinism
  - Agents should handle such uncertainty

- Previous approaches like Logic represent all possible world states

- Such approaches can't be used as multiple possible states n eed to be enumerated to handle the uncertainty in our infor mation.

Belief

- You can reach Bangalore Airport from MG Road within 90 mins if you go by route A.
- Such information can only provide a degree of belief, e.g., w e are 80% conf ident that it would be true on any given day
- In order to deal such degree of belief, we need **Probability Theory**.

Knowledge Representation

As discussed earlier the knowledge can be represent in any of the following language with respect to aspects.

Language	Ontological Commitment (What exists in the world)	Epistemological Commitment (What an agent believes about facts)
Propositional logic	facts	true/false/unknown
First-order logic	facts, objects, relations	true/false/unknown
Temporal logic	facts, objects, relations, times	true/false/unknown
Probability theory	facts	degree of belief $\in [0, 1]$
Fuzzy logic	facts with degree of truth $\in [0, 1]$	known interval value

Probability Theory

- Probability provides a way to summarize the uncertainty
- You can reach Bangalore Airport from MG Road within 90 mins if you go by route A.
  - If we know that it is evening, the Probability of the statement can be 0.4
  - Based on available information the Probability would capture the belief better.

Sample Space: Set of all possible outcomes.

- Ex: After tossing 2 coins, the set of all possible outcomes are
- {HH, HT, TH, TT}

Event: A subset of a sample space.

- An event of interest might be – {HH}.

Probability Basics – Probability Model

A fully specified **probability model** associates a numerical probability P( ) with each possible world.

The basic axioms

1. Every possible world has a probability between 0 and 1
2. Sum of probabilities of possible worlds is 1
3. E.g., P(HH) = 0.25; P(HT) = 0.25; P(TT) = 0.25, P(TH) = 0.25

$$0 \leq P(\omega) \leq 1 \text{ for every } \omega \text{ and } \sum_{\omega \in \Omega} P(\omega) = 1$$

Statistics & Probability

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- Statistics – Statistics provides a way of organizing data to extract information on a wider and objective basis than relying on personal experience.
  - Analyse the past events
- Probability – Predict the likelihood of a future event
- Questions addressed:
  - *Statistics* – How ideal is the world?
  - *Probability* – What will happen in a given ideal world?

Probability Basics – Exclusive / Exhaustive events

- Mutually Exclusive Events:
  - Two events (or propositions) are mutually exclusive or disjoint if they cannot both occur at the same time(be true).
  - A clear example is the set of outcomes of a single coin toss, which can result in either heads or tails, but not both.

- Exhaustive Events:
  - The set of events out of which one will definitely occur whenever the experiment is performed is called exhaustive events in probability. The union of the exhaustive events gives the entire sample space.

  A set of events is jointly or collectively exhaustive if at least one of the events must occur.

  E.g., when rolling a six-sided die, the events 1, 2, 3, 4, 5, and 6 are collectively exhaustive.

Probability Basics - Propositions

- Probabilistic assertions (Propositions)
  - Usually not a particular world event but about a set of them
  - E.g., two dice when rolled, a proposition can be “the sum of dice is 11”

- For any proposition ,
- P(sum = 11) = P((5, 6)) + P((6, 5))

  = 1/36 + 1/36

  = 1/18

Probability Assertions and Propositions

- Proposition – describe sets of possible worlds written in combination of Propositional Logic and Constraint Satisfaction (CSP) notation

From the concepts of CSP

- Variables: In probabilities, we call all variables as random variables. E.g., Die_1, Sum. Random variables usually begin with capital letter.
- Domain: Each random variable would have a set of values it can take. E.g,. Die_1 = {1, 2, 3, 4, 5, 6}. The values are always lowercase. For Boolean variable the domain is {true, false}
- Abbrevation: For variable A = True, is usually abbreviated as *"a"* and A = False as *"¬a"* . E.g., The use of *"cavity", "toothache"* are abbreviations.

Probability Basics – Unconditional / Prior

- Unconditional / Prior probabilities:
- Propositions like P(sum = 11) or P(two dices rolling equals) are called Unconditional or Prior probabilities

They refer to degree of belief in absence of any other information

Probability Basics - Conditional

- However, most of the time we have some information, we call it **evidence**

E.g., we can be interested in two dice rolling a double (i.e., 1,1 or 2,2, etc)

When one die has rolled 5 and the other die is still spinning

Here, we not interested in unconditional probability of rolling a double

Instead, the **conditional** or **posterior** probability for rolling a double given the first die has rolled a 5

$P(doubles \mid Die_1 = 5)$ where | is pronounced "given"

E.g., if you are going for a dentist for a checkup, P(cavity) = 0.2

- If you have a toothache, then P(cavity | toothache) = 0.6

Probability Basics - Conditional

Conditional probabilities can be expressed in unconditional probabilities

E.g., for proposition a, b Holds when P(b) > 0

$$P(a \mid b) = \frac{P(a \wedge b)}{P(b)}$$

Product rule :

$$P(a \wedge b) = P(a \mid b)P(b)$$

Probability Distribution

- If there is a random variable Weather with domain {sunny, rain, cloudy, snow}, we could write
- P(Weather = sunny) = 0.6
- P(Weather = rain) = 0.1
- P(Weather = cloudy) = 0.2
- P(Weather = snow) = 0.1

- **P**(Weather) = <0.6, 0.1, 0.2, 0.1>
- **P** defines a probability distribution for the random variable Weather
- **P** is used for conditional distributions; **P**(X|Y) gives values for **P**(X = i, | Y = j) for all combinations of i and j

Probability Density Function

- For continuous variables (e.g., weight) it is infeasible to write the distribution as a vector

- Instead, we define the probability as a function of the value the variable can take
- P(Weight = x) = Normal(x | mean = 60, std = 10)

- It says that the weight of an individual is Normal (Gaussian) distributed with average weight being 60 kgs and a deviation of 10 kgs

- They are called as Probability Density Function

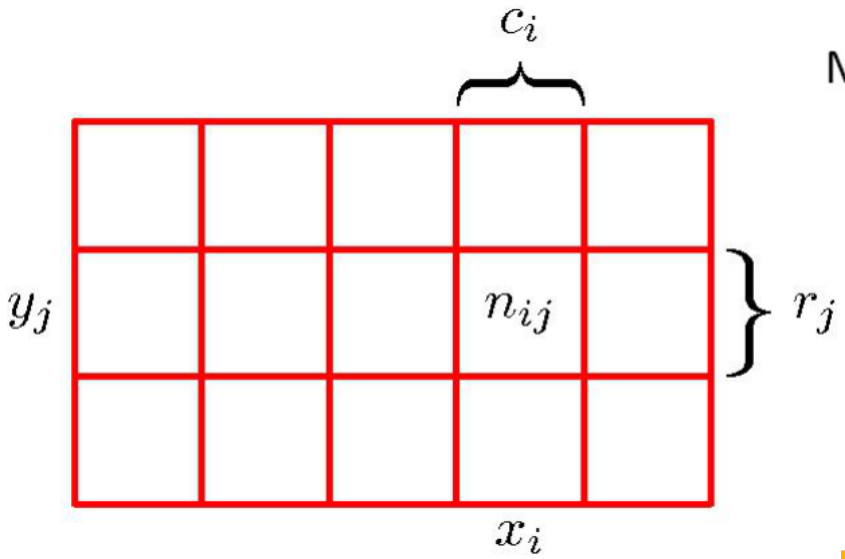
Joint Probability Distributions

- Instead of distribution over single variable, we can model distribution over multiple variables, separated by comma
- E.g., **P**(A, B) = **P**(A | B) . **P**(B)
- **P**(A, B) is the probability distribution over combination of all values of A and B
- E.g., if A = Weather and B = Cavity.

$$
\begin{aligned}
P(W = sunny \wedge C = true) &= P(W = sunny | C = true)\, P(C = true) \\
P(W = rain \wedge C = true) &= P(W = rain | C = true)\, P(C = true) \\
P(W = cloudy \wedge C = true) &= P(W = cloudy | C = true)\, P(C = true) \\
P(W = snow \wedge C = true) &= P(W = snow | C = true)\, P(C = true) \\
P(W = sunny \wedge C = false) &= P(W = sunny | C = false)\, P(C = false) \\
P(W = rain \wedge C = false) &= P(W = rain | C = false)\, P(C = false) \\
P(W = cloudy \wedge C = false) &= P(W = cloudy | C = false)\, P(C = false) \\
P(W = snow \wedge C = false) &= P(W = snow | C = false)\, P(C = false)\ .
\end{aligned}
$$

Probability Theory

- X and Y are two discrete random variables.
- X can take any of the values ***xi***, i = 1, . . . , M, and Y can take the values ***yj***, j = 1, . . . , L.
- Let a total of N trials in which we sample both of the variables X and Y, and let the number of such trials in which X = ***xi*** and Y = ***yj*** be ***nij*** .
- Let the number of trials in which X takes the value ***xi*** be denoted by ***ci***, and similarly let the number of trials in which Y takes the value ***yj*** be denoted by ***rj*** .



Marginal Probability

$$p(X = x_i) = \frac{c_i}{N}.$$

Conditional Probability

$$p(Y = y_j | X = x_i) = \frac{n_{ij}}{c_i}$$

Joint Probability

$$p(X = x_i, Y = y_j) = \frac{n_{ij}}{N}$$

Full Joint Probability Distribution

- A probability model is completely determined by the joint distribution for all of random variables – full joint probability distribution
- E.g., **P**(Catch, Cavity, Toothache)
- Can be represented by 2X2X2 table to represent all combinations of the variables with values corresponding to the probabilities of seeing those combinations respectively.

Probabilistic Inference

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- Computation of posterior probabilities given observed evidence, i.e., full joint probability distribution

	toothache		¬*toothache*	
	catch	¬*catch*	*catch*	¬*catch*
cavity	0.108	0.012	0.072	0.008
¬*cavity*	0.016	0.064	0.144	0.576

Query: P(cavity ∨ toothache)

0.108 + 0.012 + 0.072 + 0.008 + 0.016 + 0.064 = 0.28

Probabilistic Inference: Marginalization

- Query: P(cavity)
- P(cavity) = 0.108 + 0.012 + 0.072 + 0.008 = 0.2

- This process is called **marginalization** or **summing out –** as we sum up probabilities for each possible value of other variable so that we take them out of equation**.**

$$\mathbf{P}(\mathbf{Y}) = \sum_{\mathbf{z} \in \mathbf{Z}} \mathbf{P}(\mathbf{Y}, \mathbf{z})$$

Probability Basics – Marginal Probability

- Marginal Probability: The probability of an event occurring P(A).
  It may be thought of as an unconditional probability.
  It is not conditioned on another event.

- Ex: From a deck of cards, the probability that a card drawn is red
  p(red) = 0.5

  From a deck of cards, the probability that a card drawn is a 4

  p(four)=1/13

Independence

- If we have two random variables, TimeToBnlrAirport and HyderabadWeather
  P(TimeToBnlrAirport, HyderabadWeather)
- To determine their relation, use the product rule
- = P(TimeToBnlrAirport | HyderabadWeather) / P(HyderabadWeather)
- However, we would argue that HyderabadWeather and TimeToBnlrAirport doesn't have any relation and hence
- P(TimeToBnlrAirport | HyderabadWeather) = P(TimeToBnlrAirport)
- This is called Independence or Marginal Independence
- Independence between propositions a and b can be written as

$$P(a \mid b) = P(a) \quad \text{or} \quad P(b \mid a) = P(b) \quad \text{or} \quad P(a \wedge b) = P(a)P(b)$$

Bayes Rule

- Using the product rule for propositions a and b

$$P(a \wedge b) = P(a \mid b)P(b) \qquad \text{and} \qquad P(a \wedge b) = P(b \mid a)P(a)$$

- Equating the right hand sides and dividing by P(a)

$$P(b \mid a) = \frac{P(a \mid b)P(b)}{P(a)}$$

- This is called the Bayes Rule.

Conditional Independence

- 2 random variables A and B are conditionally independent given C iff

  $P(a, b \mid c) = P(a \mid c)\, P(b \mid c)$ for all values a, b, c

- More intuitive (equivalent) conditional formulation
  - A and B are conditionally independent given C iff

    $P(a \mid b, c) = P(a \mid c)$ OR $P(b \mid a, c) = P(b \mid c)$, for all values a, b, c

  - Intuitive interpretation:

$P(a \mid b, c) = P(a \mid c)$ tells us that learning about b, given that we already know c, provides no change in our probability for a, i.e., b c ontains no information about a beyond what c provides.

Conditional Independence

- Can generalize to more than 2 random variables
  - E.g., K different symptom variables X1, X2, … XK, and C = disease
  - P(X1, X2,…. XK | C) = P(Xi | C)
  - Also known as the naïve Bayes assumption

Bayesian Networks

- In Probabilities, we have seen that random variables can be related with joint distribution, conditionality, independence, etc.

  However, such relations are def ined in a formal semantics using a network topology and probabilities

- Bayesian Networks is a field that represents such relations systematically

- We explore the use of probability theory to objects and relations (like in first-order logic)

Bayesian Networks

- Full joint probability distributions can be used for inference however can become intractable as the number of variables grow

- Bayesian Networks represent dependencies among variables

  - Can represent full joint probability distribution

What is a Bayesian Network

A Bayesian network is a directed graph in which each node is annotated with quantitative probability information.

- Each node corresponds to a random variable, which may be discrete or continuous.
- A set of directed links or arrows connects pairs of nodes. If there is an arrow from node X to node Y , X is said to be a parent of Y. The graph has no directed cycles (and hence is a directed acyclic graph, or *DAG*).
- Each node Xi has a conditional probability distribution P(Xi | Parents(Xi)) that quantifier the effect of the parents on the node.

Bayesian Networks

Once the topology of the Bayesian Network is laid out,

- Will specify the conditional probability distribution for each variable given its parents
- Such that the topology and conditional probabilities suffices to specify the full joint probability distribution of all variables

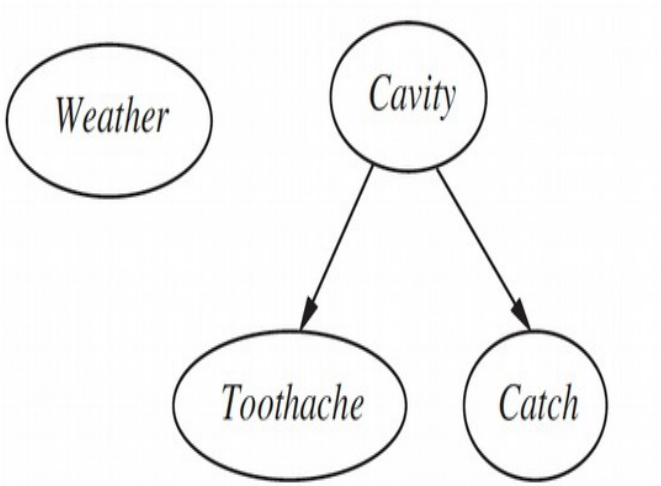
Example Bayesian Net

A simple world with four randomvariables

Weather, Toothache, Cavity, Catch

Weather is independent of other variables

Toothache and Catch are conditionally independe
given Cavity



P(Toothache, Catch | Cavity) = P(Too thache | Cavity) .
P(Catch | Cavity)

Cavity is a direct cause of Toothache and Catch

No direct relation between Toothache and Catch exists.

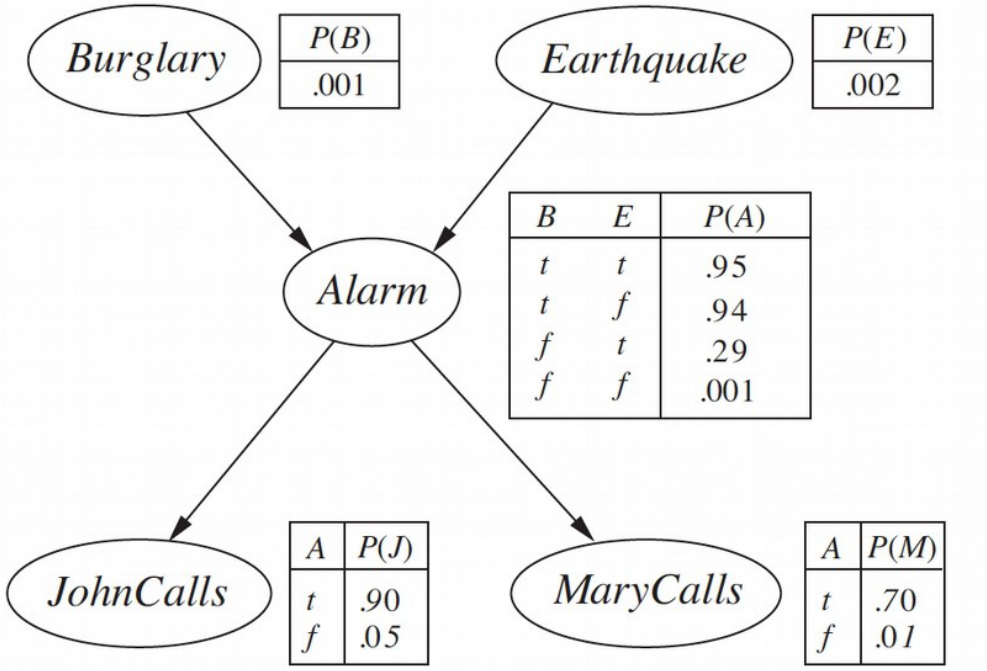
Example Bayesian Net

A Burglary Alarm System

- Fairly reliable on detecting a burglary Also responds to earthquakes
- Two neighbors John and Mary are asked to call you at workwhen Burglary happens and they hear the Alarm
- need to estimate the probability of a burglary
- John nearly always calls when he hears the alarm, however some times confuses the telephone ring with alarm and calls then too
  - Mary like loud music and often misses the alarm altogether Problem: Given the information that who has / has not called we need to estimate the probability of a burglary

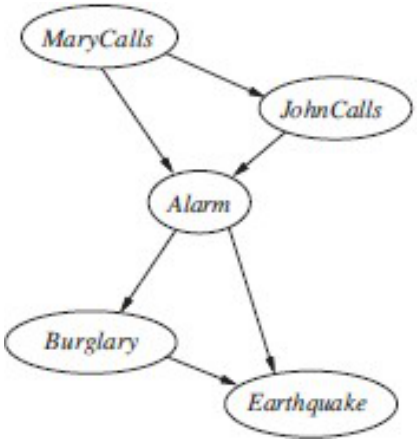
Example Bayes Net: Burglary Alarm

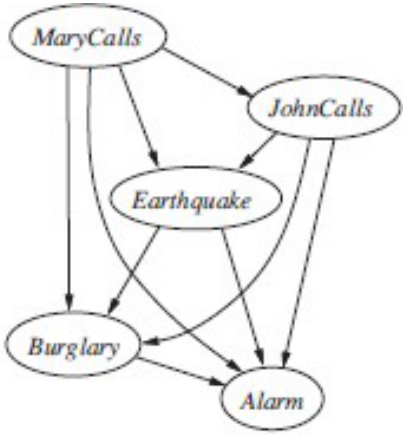
- Conditional distributions shown are Conditional Probability Tables (CPT)
- Used for discrete variables



Not so good Bayes Nets: Burglary Alarm

- We will get a compact Bayesian network only if we choose the node ordering well.
- What happens if we happen to choose the wrong order? Suppose
- we decide to add the nodes in the order
- MaryCalls, JohnCalls, Alarm, Burglary, Earthquake or
- MaryCalls, JohnCalls, Earthquake , Burglary , Alarm



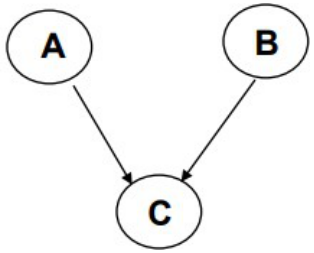


Network structure depends on order of introduction. In each network, nodes are introduced nodes top-to-bottom order.
The network has more links that represent tenuous relationships and require diff icult and unnatural probability judgments

Semantics of Bayesian Nets

- Captures Full Joint Distribution of all variables given the topology with conditional distributions
- Example: Simple bayesian network with 3 variables.

p(A,B,C) = p(C|A,B)p(A)p(B) ⟷



A generic entry in joint distribution is a probability of conjunction of particular assignments to each variable such as

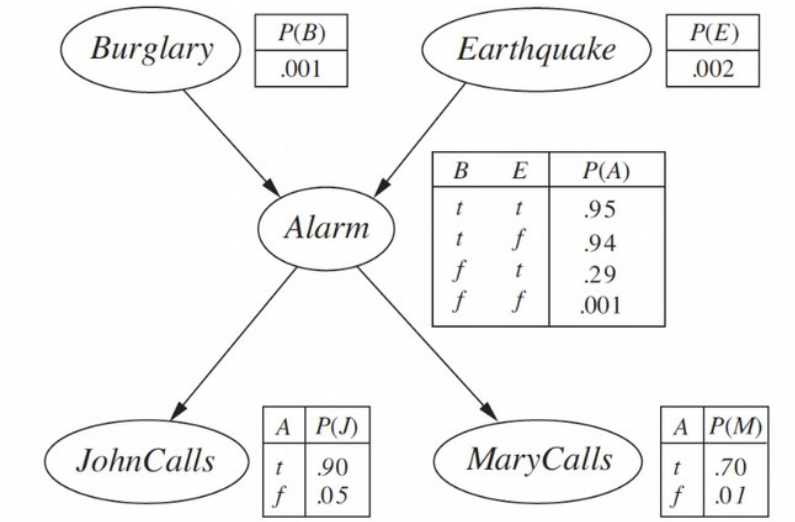
$$P(X_1 = x_1 \wedge X_2 = x_2 \wedge X_3 = x_3 \; .. \wedge X_n = x_n)$$

This can be rewritten using conditional probabilities as

$$P(x_1, \ldots, x_n) = \prod_{i=1}^{n} P(x_i \,|\, parents(X_i))$$

Example

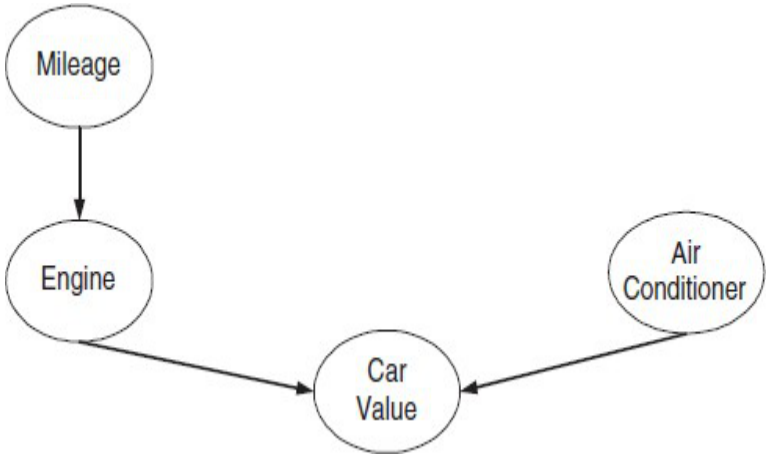
· Calculate the probability that alarm has sounded, but neither burglary nor earthquake happened, and both John and Mary called.



$$P(j, m, a, \neg b, \neg e) = P(j \mid a)P(m \mid a)P(a \mid \neg b \wedge \neg e)P(\neg b)P(\neg e)$$
$$= 0.90 \times 0.70 \times 0.001 \times 0.999 \times 0.998 = 0.000628$$

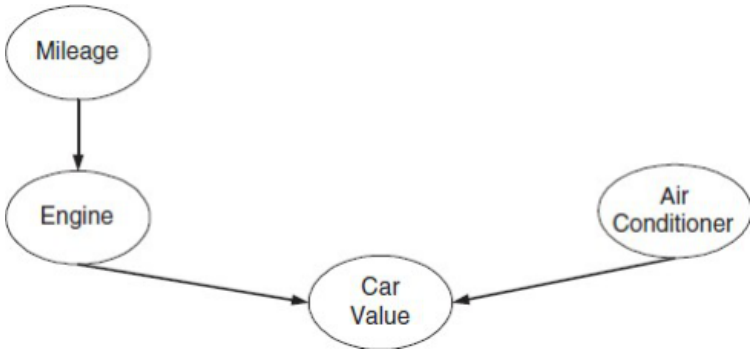
Example

- Given the Bayesian Belief Network and a set of observations, compute probabilities associated with each node.



Mileage	Engine	Air Conditioner	Number of Records with Car Value=Hi	Number of Records with Car Value=Lo
Hi	Good	Working	3	4
Hi	Good	Broken	1	2
Hi	Bad	Working	1	5
Hi	Bad	Broken	0	4
Lo	Good	Working	9	0
Lo	Good	Broken	5	1
Lo	Bad	Working	1	2
Lo	Bad	Broken	0	2

Example



Mileage	Engine	Air Conditioner	Number of Records with Car Value=Hi	Number of Records with Car Value=Lo
Hi	Good	Working	3	4
Hi	Good	Broken	1	2
Hi	Bad	Working	1	5
Hi	Bad	Broken	0	4
Lo	Good	Working	9	0
Lo	Good	Broken	5	1
Lo	Bad	Working	1	2
Lo	Bad	Broken	0	2

- P(Mileage=Hi) = 0.5
- P(Air Cond=Working) = 0.625
- P(Engine=Good|Mileage=Hi) = 0.5
- P(Engine=Good|Mileage=Lo) = 0.75

P(Value=High|Engine=Good, Air Cond=Working) = 0.750
P(Value=High|Engine=Good, Air Cond=Broken) = 0.667
P(Value=High|Engine=Bad, Air Cond=Working) = 0.222
P(Value=High|Engine=Bad, Air Cond=Broken) = 0

Probabilistic Reasoning over Time
Hidden Markov Model

Partial Observability

- Agents in partially observable environment should keep a track of current state to the extent allowed by sensors
  - E.g., Robot moving in a new maze
- Agent maintains a **belief state** representing the current possible world states
- **Transition Model:** Using belief state and transition model, the agent can how the world might evolve in next time step
- **Sensor Model:** With the observed percepts and sensor model, the agent can update the belief state.

Degree of belief

Earlier, the states were captured as

- Facts in Logic which can take either True/False

- To capture the degree of belief we will use Probability Theory.

- We model the change in world using a variable for each aspect of state and at each point in time

- **Transition models** – describe the probability distribution of variables at time t, given the state of the world at past times.

- **Sensor models** – describe the probability of each percept at time 't', given the current state of the world.

Time and Uncertainty

- Static World: Each random variable would have a single fixed value
  - E.g., Diagnosing a broken car

- Dynamic World: The state information keeps changing with time
  - E.g., treating a diabetic patient, tracking the location of robot, tracking economic activity of a nation etc..

States and Observations

- Time slices: World is observed in time slices. Each slice has a set of random variables, some observable and some not.

- Assumption: We will assume same subset of random variables are observable in each time slice

- Denoted by
- **Et** – set of observable random variables at time t
- **Xt** – set of unobserved random variables at time t

Example

- A security guard at a secret underground station
- He couldn't access the outside world so has no idea of weather

- He wants to know whether it is raining
- Only observed variable: His director coming in with/without an umbrella

- For each day t, Et - single evidence variable Umbrella t(Ut) (Y/N)

  Xt – single state variable Raint (Rt) (Yes/No)

Formulation

- Assumption: The time interval between slices is fixed
- Also, state variables start at t = 0
- Evidence variables start at t = 1

- Umbrella world can be represented by
  - State variables – R0, R1, R2, R3, …
  - Evidence variables – U1, U2, U3, …
  - Notation: Time slice a:b → Xa:b → Xa to Xb

Transition and Markov models

- Transition Model: Probability distribution over latent(hidden) state variables given the previous values, P(Xt | X0:t-1)
- X0:t-1 – unbounded in size as t increases
- Markov Assumption: Current state depends only on finite fixed number of previous states

- Markov Process/Chain – Processes satisfying this assumption

First Order Markov Process

- When the current state depends only on the previous state but not on earlier states

$$P(X_t \mid X_{0:t-1}) = P(X_t \mid X_{t-1})$$

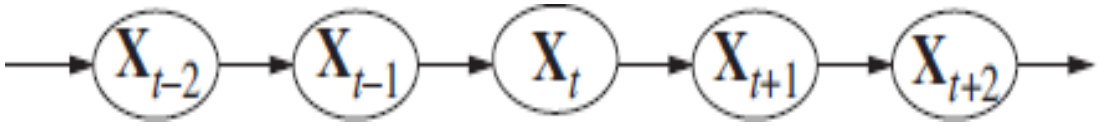
- Transition Model: $P(X_t \mid X_{t-1})$

- Second-order Markov Process:

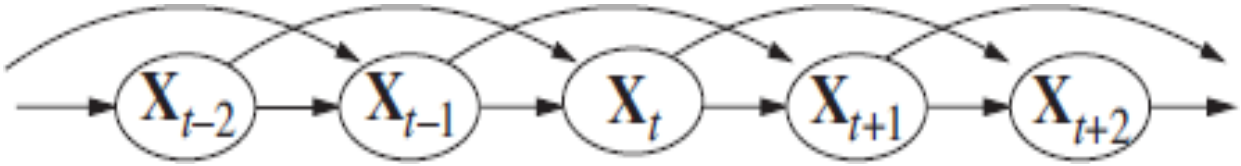
$$P(X_t \mid X_{0:t-1}) = P(X_t \mid X_{t-2}, X_{t-1}).$$

Bayesian Network

First-order Markov process



Second-order Markov process



Stationary Process

- Given the inf inite values of t, the probability distribution governing $P(X_t | X_{t-1})$

- Assumption: Changes in the world states are caused by a stationary process – a process of change that are governed by laws that do not themselves change over time

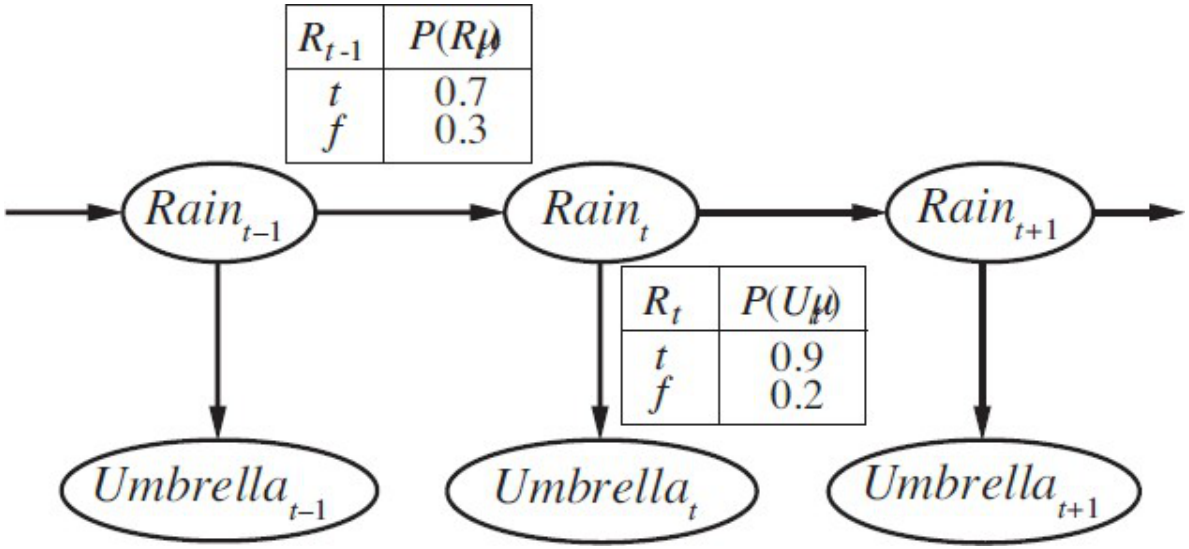
- Therefore, $P(R_t | R_{t-1})$ is same for all t
- One conditional probability table

$P(R_{t-1})$	$P(R_t \mid R_{t-1})$
True	0.7
False	0.3

Sensor Model

- Evidence Variable Et can depend on previous variables as well as current state variables, i.e., $P(E_t | X_{0:t}, E_{0:t-1})$
- Sensor Markov Assumption: $P(E_t | X_{0:t}, E_{0:t-1}) = P(E_t | X_t)$
- Sensor Model: $P(E_t | X_t)$

Bayesian Net for Umbrella Example



- Bayesian network structure and conditional distributions describing the umbrella world.
- The transition model is $P(Rain_t \mid Rain_{t-1})$ and the sensor model is $P(Umbrella_t \mid Rain_t)$.

Inference in Temporal Models

1. Filtering – computing the belief state – posterior distribution of the most recent state – given all evidence to date
   - $P(X_t \mid E_{1:t})$
2. Predicting – Computing the posterior distribution over future state given all evidence to date
   - $P(X_{t+k} \mid E_{1:t})$ where $k > 0$
3. Smoothing – Computing the posterior distribution over past state given all evidence to date
   - $P(X_k \mid E_{1:t})$ where $0 \leq k < t$
4. Most likely sequence – Given a sequence of observations, find the sequence of states that is most likely to have generated the observations
   - $\text{argmax}_{x1:t} P(x_{1:t} \mid e_{1:t})$
5. Learning – Learning the transition and sensor models. Can be learned from examples
   - $T = P(X_t \mid X_{t-1})$
   - $E = P(E_t \mid X_t)$

Inference - Filtering

- Recursive Elimination

$$\mathbf{P}(\mathbf{X}_{t+1} \mid \mathbf{e}_{1:t+1}) = f(\mathbf{e}_{t+1}, \mathbf{P}(\mathbf{X}_t \mid \mathbf{e}_{1:t}))$$

Inference - Filtering

$$\begin{aligned}
\mathbf{P}(\mathbf{X}_{t+1} \mid \mathbf{e}_{1:t+1}) &= \mathbf{P}(\mathbf{X}_{t+1} \mid \mathbf{e}_{1:t}, \mathbf{e}_{t+1}) \quad \text{(dividing up the evidence)} \\
&= \alpha\, \mathbf{P}(\mathbf{e}_{t+1} \mid \mathbf{X}_{t+1}, \mathbf{e}_{1:t})\, \mathbf{P}(\mathbf{X}_{t+1} \mid \mathbf{e}_{1:t}) \quad \text{(using Bayes' rule)} \\
&= \alpha\, \mathbf{P}(\mathbf{e}_{t+1} \mid \mathbf{X}_{t+1})\, \mathbf{P}(\mathbf{X}_{t+1} \mid \mathbf{e}_{1:t}) \quad \text{(by the sensor Markov assumption)}
\end{aligned}$$

α is a normalizing constant used to make probabilities sum up to 1.

$$\begin{aligned}
\mathbf{P}(\mathbf{X}_{t+1} \mid \mathbf{e}_{1:t+1}) &= \alpha\, \mathbf{P}(\mathbf{e}_{t+1} \mid \mathbf{X}_{t+1}) \sum_{\mathbf{x}_t} \mathbf{P}(\mathbf{X}_{t+1} \mid \mathbf{x}_t, \mathbf{e}_{1:t}) P(\mathbf{x}_t \mid \mathbf{e}_{1:t}) \\
&= \alpha\, \mathbf{P}(\mathbf{e}_{t+1} \mid \mathbf{X}_{t+1}) \sum_{\mathbf{x}_t} \mathbf{P}(\mathbf{X}_{t+1} \mid \mathbf{x}_t) P(\mathbf{x}_t \mid \mathbf{e}_{1:t}) \quad \text{(Markov assumption)}
\end{aligned}$$

We can think of the f iltered estimate $\mathbf{P}(\mathbf{X}_t \mid \mathbf{e}_{1:t})$ as a "message" $\mathbf{f}_{1:t}$ that is propagated forward along the sequence, modif ied by each transition and updated by each new observation.

- Forward Message → $f_{1:t} = P(X_t \mid e_{1:t})$

$$\mathbf{f}_{1:t+1} = \alpha\, \text{FORWARD}(\mathbf{f}_{1:t}, \mathbf{e}_{t+1})$$

Filtering Example

- Compute P(R2 | u1:2)

- Transition Model = P(Rt | Rt-1)

Rt-1	P(Rt|Rt-1)
True	0.7
False	0.3

- Sensor Model = P(Et| Xt)

Rt	P(Et|Rt)
True	0.9
False	0.2

Filtering Example

- Day 0: No observations, only the prior beliefs of security guard: $P(R_0) = \langle 0.5, 0.5 \rangle$

- Day 1: Umbrella appears U1 = True

$$\mathbf{P}(R_1) = \sum_{r_0} \mathbf{P}(R_1 \mid r_0) P(r_0)$$
$$= \langle 0.7, 0.3 \rangle \times 0.5 + \langle 0.3, 0.7 \rangle \times 0.5 = \langle 0.5, 0.5 \rangle$$

  - Adding the evidence U1

$$\mathbf{P}(R_1 \mid u_1) = \alpha \mathbf{P}(u_1 \mid R_1) \mathbf{P}(R_1) = \alpha \langle 0.9, 0.2 \rangle \langle 0.5, 0.5 \rangle$$
$$= \alpha \langle 0.45, 0.1 \rangle \approx \langle 0.818, 0.182 \rangle .$$

Filtering Example

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- Day 2: the umbrella appears U2 = True

$$\begin{aligned}\mathbf{P}(R_2 \mid u_1) &= \sum_{r_1} \mathbf{P}(R_2 \mid r_1) P(r_1 \mid u_1) \\ &= \langle 0.7, 0.3 \rangle \times 0.818 + \langle 0.3, 0.7 \rangle \times 0.182 \approx \langle 0.627, 0.373 \rangle\end{aligned}$$

  - Then adding the evidence U2

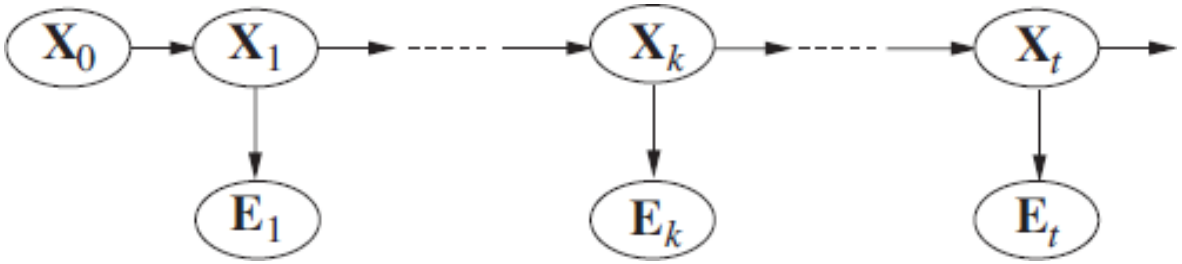
$$\begin{aligned}\mathbf{P}(R_2 \mid u_1, u_2) &= \alpha \mathbf{P}(u_2 \mid R_2) \mathbf{P}(R_2 \mid u_1) = \alpha \langle 0.9, 0.2 \rangle \langle 0.627, 0.373 \rangle \\ &= \alpha \langle 0.565, 0.075 \rangle \approx \langle 0.883, 0.117 \rangle \, .\end{aligned}$$

Inference: Prediction

- Prediction can be seen as f iltering without the additi on of new evidence

$$\mathbf{P}(\mathbf{X}_{t+k+1} \mid \mathbf{e}_{1:t}) = \sum_{\mathbf{x}_{t+k}} \mathbf{P}(\mathbf{X}_{t+k+1} \mid \mathbf{x}_{t+k}) P(\mathbf{x}_{t+k} \mid \mathbf{e}_{1:t})$$

Inference: Smoothing



- Smoothing – Process of computing the distribution over past states given evidence up to present, i.e., $P(X_k \mid e_{1:t})$ $0 \leq k < t$

  Split the computation into two parts, evidence up to k, and evidence from k+1 to t

$$
\begin{aligned}
\mathbf{P}(\mathbf{X}_k \mid \mathbf{e}_{1:t}) &= \mathbf{P}(\mathbf{X}_k \mid \mathbf{e}_{1:k}, \mathbf{e}_{k+1:t}) \\
&= \alpha\, \mathbf{P}(\mathbf{X}_k \mid \mathbf{e}_{1:k}) \mathbf{P}(\mathbf{e}_{k+1:t} \mid \mathbf{X}_k, \mathbf{e}_{1:k}) \quad \text{(using Bayes' rule)} \\
&= \alpha\, \mathbf{P}(\mathbf{X}_k \mid \mathbf{e}_{1:k}) \mathbf{P}(\mathbf{e}_{k+1:t} \mid \mathbf{X}_k) \quad \text{(using conditional independence)} \\
&= \alpha\, \mathbf{f}_{1:k} \times \mathbf{b}_{k+1:t} \,.
\end{aligned}
$$

Inference Smoothing

- Backward Message → $b_{k+1:t} = P(e_{k+1:t} \mid X_k)$

$$\begin{aligned}
\mathbf{P}(\mathbf{e}_{k+1:t} \mid \mathbf{X}_k) &= \sum_{\mathbf{x}_{k+1}} \mathbf{P}(\mathbf{e}_{k+1:t} \mid \mathbf{X}_k, \mathbf{x}_{k+1}) \mathbf{P}(\mathbf{x}_{k+1} \mid \mathbf{X}_k) \quad \text{(conditioning on } \mathbf{X}_{k+1}\text{)} \\
&= \sum_{\mathbf{x}_{k+1}} P(\mathbf{e}_{k+1:t} \mid \mathbf{x}_{k+1}) \mathbf{P}(\mathbf{x}_{k+1} \mid \mathbf{X}_k) \quad \text{(by conditional independence)} \\
&= \sum_{\mathbf{x}_{k+1}} P(\mathbf{e}_{k+1}, \mathbf{e}_{k+2:t} \mid \mathbf{x}_{k+1}) \mathbf{P}(\mathbf{x}_{k+1} \mid \mathbf{X}_k) \\
&= \sum_{\mathbf{x}_{k+1}} P(\mathbf{e}_{k+1} \mid \mathbf{x}_{k+1}) P(\mathbf{e}_{k+2:t} \mid \mathbf{x}_{k+1}) \mathbf{P}(\mathbf{x}_{k+1} \mid \mathbf{X}_k)
\end{aligned}$$

$$\mathbf{b}_{k+1:t} = \text{BACKWARD}(\mathbf{b}_{k+2:t}, \mathbf{e}_{k+1})$$

Smoothing Example

- Compute $P(R_1 \mid u_{1:2})$

$$\mathbf{P}(R_1 \mid u_1, u_2) = \alpha\, \mathbf{P}(R_1 \mid u_1)\, \mathbf{P}(u_2 \mid R_1)$$

- We already know $P(R_1 \mid u_1) = \langle 0.818, .182 \rangle$

$$\begin{aligned}\mathbf{P}(u_2 \mid R_1) &= \sum_{r_2} P(u_2 \mid r_2) P(\ \mid r_2) \mathbf{P}(r_2 \mid R_1) \\ &= (0.9 \times 1 \times \langle 0.7, 0.3 \rangle) + (0.2 \times 1 \times \langle 0.3, 0.7 \rangle) = \langle 0.69, 0.41 \rangle\end{aligned}$$

- Plugging it back in

$$\mathbf{P}(R_1 \mid u_1, u_2) = \alpha\, \langle 0.818, 0.182 \rangle \times \langle 0.69, 0.41 \rangle \approx \langle 0.883, 0.117 \rangle$$

Forward Backward Algorithm

function FORWARD-BACKWARD(**ev**, $prior$) **returns** a vector of probability distributions
  inputs: **ev**, a vector of evidence values for steps $1, \ldots, t$
    $prior$, the prior distribution on the initial state, $\mathbf{P}(\mathbf{X}_0)$
  local variables: **fv**, a vector of forward messages for steps $0, \ldots, t$
    b, a representation of the backward message, initially all 1s
    sv, a vector of smoothed estimates for steps $1, \ldots, t$

  $\mathbf{fv}[0] \leftarrow prior$
  for $i = 1$ **to** t **do**
    $\mathbf{fv}[i] \leftarrow$ FORWARD$(\mathbf{fv}[i-1], \mathbf{ev}[i])$
  for $i = t$ **downto** 1 **do**
    $\mathbf{sv}[i] \leftarrow$ NORMALIZE$(\mathbf{fv}[i] \times \mathbf{b})$
    $\mathbf{b} \leftarrow$ BACKWARD$(\mathbf{b}, \mathbf{ev}[i])$
  return sv

Most Likely Sequence

- Suppose the umbrella sequence is [true, true, false, true, true] What
- is the likely weather sequence that can explain this?

2^5 possible sequences

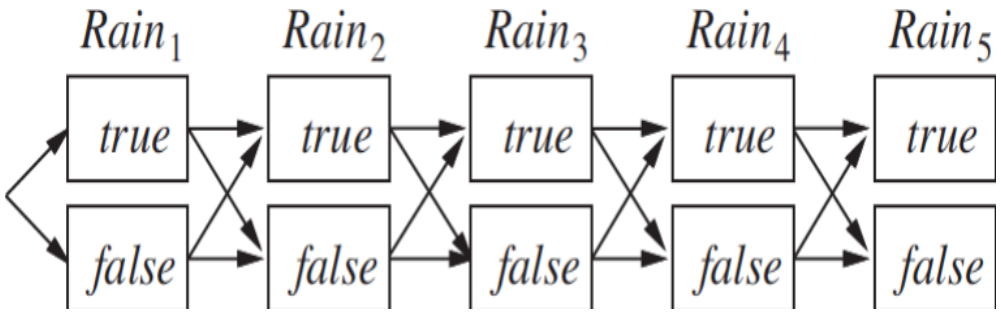
- In NLP

The Observed sequence are the words in a sentence:
[Ram, killed, Ravan]

The hidden sequence are the POS tags
[NNP, VBD, NNP]

Most Likely Sequence

- Algorithm
  - Linear Time
  - View each sequence as a path through a graph whose nodes are possible states at each step



  - Finding the most likely path – Product of transition probabilities along the path and the probabilities of given observations at each state

Most Likely Sequence

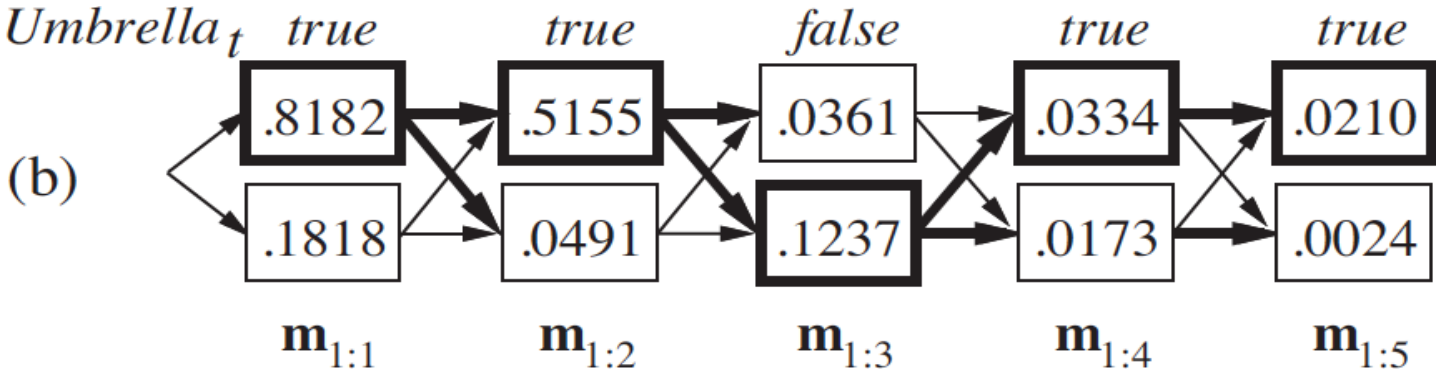
- Algorithm

  - The most likely path to say Rain5 = True consists of most likely path to some state at time 4 followed by transition to Rain5 = True
  - Recursive relation

$$\max_{\mathbf{x}_1 \ldots \mathbf{x}_t} \mathbf{P}(\mathbf{x}_1, \ldots, \mathbf{x}_t, \mathbf{X}_{t+1} \mid \mathbf{e}_{1:t+1})$$

$$= \alpha \mathbf{P}(\mathbf{e}_{t+1} \mid \mathbf{X}_{t+1}) \max_{\mathbf{x}_t} \left(\mathbf{P}(\mathbf{X}_{t+1} \mid \mathbf{x}_t) \max_{\mathbf{x}_1 \ldots \mathbf{x}_{t-1}} P(\mathbf{x}_1, \ldots, \mathbf{x}_{t-1}, \mathbf{x}_t \mid \mathbf{e}_{1:t}) \right)$$

Most Likely Sequence



- In order to f ind the actual sequence, the algorithm need to record, for each state, the best state that leads to it; indicated by bold arrows
- The optimal sequence is identif ied by following bold arrows backwards from the best f inal state
- This is called **Viterbi algorithm**

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Questions???

Thank you.